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MIL-C-50785/12A (AR)
27 November 1992
SUPERSEDING
MIL-C-50785/12 (MU)
24 October 1975

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 12000012

(COMPUTATION CONTROL)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50785.

REQUIREMENTS

1 Item definition. This assembly is a single-ended printed circuit board consisting of a triangle wave generator and two inverters. The assembly is used to invert and/or amplify voltages within the Sight Current Generator Draawing Number 12000239 in which it is assembled.

2 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000012.

3 Government furnished property list. Not applicable.

4 Characteristics.


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4.1 Performance. The assembly shall be capable of performing functions a through c when assembled in accordance with Drawing Number 12000012.

a. Inverter (Q1) shall invert one of the inputs required to control the "Ready to Fire" circuitry and "Acquisition Time Delay" circuitry.

b. Inverter (Q2) shall invert one of the inputs required to control the "Ready to Fire" circuitry.

c. The triangle wave generator (AR - AR2) shall generate an 18 volt P-P triangular wave within a period of approximately 0.67 millisecond.

4.2 Input and output levels.

4.3 Input power. Input power shall be supplied to the assembly as specified in Table I.

TABLE I - INPUT VOLTAGES

| Pin | Voltage |
|---|---|
| 4 and 25 | +15 Vdc ± 2.0 percent |
| 14 | -15 Vdc ± 2.0 percent |
| 1 and 28 | Ground |

4.4 Output levels. The requirements specified in 4.5 through 4.7 shall be met when supplied with the specified voltage input.

4.5 Pin 26 output. The output shall be +15 Vdc ±2.0 percent between pin 26 and pin 28 with zero Vdc input applied between pin 18 and pin 28, and 0.5 ± Vdc with +15 Vdc input applied between pin 18 and pin 28.

4.6 Pin 27 output. The output between pin 27 and pin 28 shall be as specified in a through c.

a. As shown in Figure 1 when +15 Vdc is removed from pin 17.

b. +15 Vdc ± 4.0 percent with zero Vdc applied to pin 17.

c. +1.0 ± 0.5 Vdc with +15 Vdc applied between pin 17 and pin 28.



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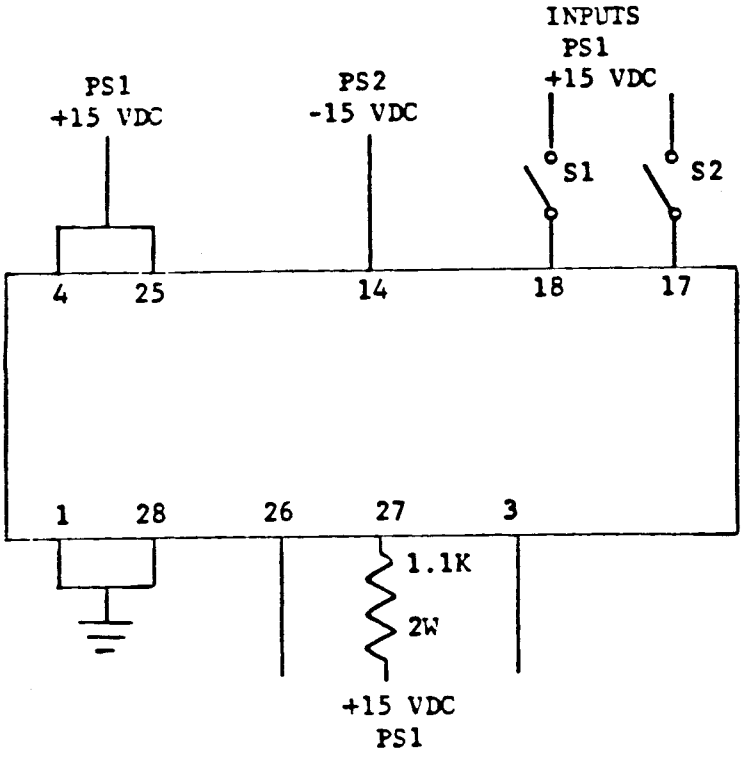



Figure 1.

4.7 Pin 3 output. The output shall be a triangular wave with an amplitude from -8.08 ± 1.08 Vdc to +8.08 ± 1.03 Vdc within a period of 0.67 ± 0.14 milliseconds.

4.8 Physical characteristics. The physical characteristics shall be as specified in drawing 12000012 and the engineering parts list.

5 Test equipment. The test equipment listed below shall be used in the performance of the applicable test methods and procedures as specified in 6.

5.1 Power supply (PS1). The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.



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5.2 Power supply (PS2). The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.3 Switches. Two single pole single throw switches are required.

5.4 Oscilloscope. The oscilloscope shall have voltages, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volt/cm to 5.0 volts/cm.

5.5 Digital voltmeter (DMV). The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

5.6 Resistor. One resistor is required with a rating of 1100 ohms ± 5.0 percent, 2.0 watts.

6 Functional test.

6.1 Test connections. Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 2 using receptacle 10545455.

6.2 Test procedure. Compliance with the requirements of 4.3 through 4.7 shall be verified as follows:

a. With switch S1 off, the voltage at pin 26 measured with the DVM shall be +15 Vdc ± 2.0 percent.

b. With switch S1 on, the voltage at pin 26 measured with the DVM shall be 0.5 ± 0.5 Vdc.

c. With switch S2 off the voltage at pin 27 measured with the DVM shall be +15 Vdc ± 4.0 percent.

d. With switch S2 on, the voltage at pin 27 measured with the DVM shall be +1.0 ± 0.5 Vdc.

e. When the switch S2 is positioned from on to off, the voltage at pin 27 measured with the oscilliscope shall be as shown in Figure 1.

f. With the oscilliscope connected to pin 3, the voltage shall be a triangular wave with an amplitude from -8.08± 1.08 Vdc to +8.08 ± 1.08 Vdc within a period of 0.67 ± 0.14 milliseconds.



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7.0 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

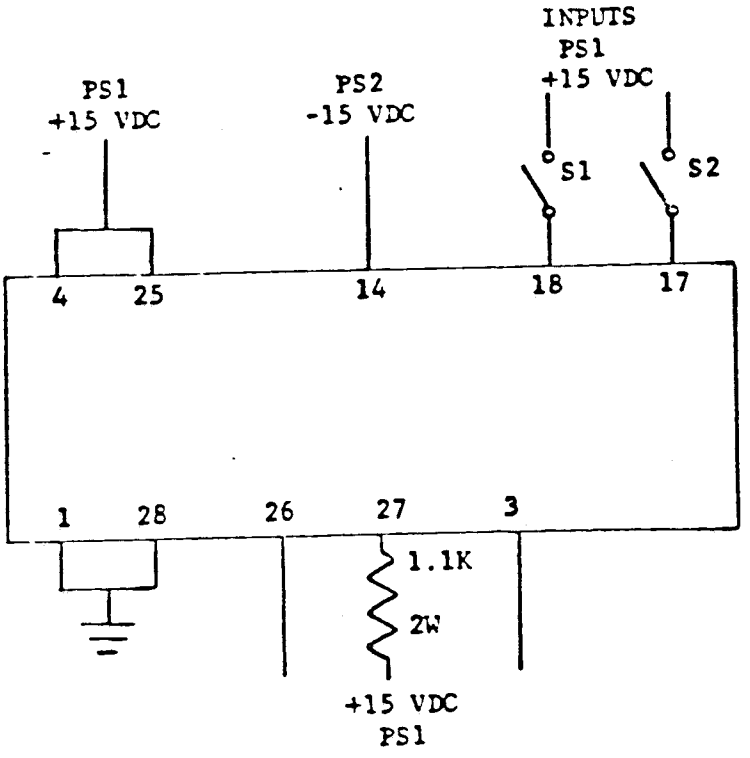



Figure 2. Test Connections

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Preparing Activity

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER MIL-C-50785/12A | 2. DOCUMENT DATE (YYMMDD) 27 NOVEMBER 1992 |
|---|---|---|

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY: 12000012 (COMPUTATION CONTROL)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
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